Mentorship, Interrupted: How GenAI Disrupts the GSoC Pipeline and Why Mentors Matter More — Appendix

I. Exploratory Analysis of Mailing-List Discussions

Because the GSoC mentors' mailing list is restricted-access, we withhold message text, thread titles, and identities to protect the privacy of the individuals involved. Supplemental Table I instead reports the anonymized thread numbers and the number of coded excerpts supporting each theme.

TABLE I: Provisional themes from the exploratory mailing-list analysis and coded excerpt counts by thread.

Theme	Description	Supporting excerpts by thread
GenAI impacts on GSoC sustainability	How GenAI affects newcomers' learning and participation, trust, and mentors' capacity and willingness to keep supporting OSS communities. **Interview focus**: Mentor effort, learning, and perceived GSoC contributions (RQ1).	ML1: 1 excerpt; ML3: 7 excerpts; ML5: 1 excerpt; ML6: 1 excerpt; ML7: 13 excerpts; ML8: 2 excerpts; ML9: 1 excerpt; ML10: 1 excerpt; ML11: 1 excerpt; ML12: 1 excerpt; ML14: 24 excerpts; ML15: 2 excerpts. **[Total: 55 / 292 excerpts]**
GenAI manifestations in GSoC	Where GenAI shows up in applications, proposals, code, and messages, including low-effort submissions and fabricated content. **Interview focus**: Changes in onboarding, interactions, and contribution review (RQ1).	ML2: 4 excerpts; ML3: 5 excerpts; ML4: 1 excerpt; ML6: 7 excerpts; ML10: 3 excerpts; ML13: 2 excerpts; ML14: 27 excerpts; ML15: 8 excerpts. **[Total: 57 / 292 excerpts]**
Stance for GenAI use	Views on whether and when GenAI use is acceptable, and why. **Interview focus**: Reasons for GenAI boundaries and how views change (RQ2).	ML1: 6 excerpts; ML2: 5 excerpts; ML3: 4 excerpts; ML6: 4 excerpts; ML7: 7 excerpts; ML9: 1 excerpt; ML10: 2 excerpts; ML11: 1 excerpt; ML14: 6 excerpts; ML15: 1 excerpt. **[Total: 37 / 292 excerpts]**
GenAI detection and verification	Efforts to spot GenAI use and check contributors' understanding, and the uncertainty and risk of wrongly attributing work to GenAI. **Interview focus**: Changes in selection and review, verification effort, and response choices (RQ1/RQ2).	ML1: 3 excerpts; ML2: 3 excerpts; ML3: 5 excerpts; ML5: 1 excerpt; ML6: 4 excerpts; ML7: 1 excerpt; ML12: 1 excerpt; ML14: 5 excerpts; ML15: 2 excerpts. **[Total: 25 / 292 excerpts]**
Response and mitigation	What mentors and organizations do or propose to handle GenAI problems, adapt assessment and mentoring, and use GenAI as support. **Interview focus**: Enacted and planned responses, their rationale, and local constraints (RQ2).	ML1: 1 excerpt; ML2: 6 excerpts; ML3: 8 excerpts; ML4: 3 excerpts; ML5: 1 excerpt; ML6: 10 excerpts; ML7: 10 excerpts; ML8: 3 excerpts; ML9: 3 excerpts; ML10: 1 excerpt; ML11: 1 excerpt; ML13: 2 excerpts; ML14: 33 excerpts. **[Total: 82 / 292 excerpts]**
Potential governance and policy gaps	Open questions about who is responsible, which rules apply, and what support is needed regarding GenAI use in GSoC, including the limits of local responses and tensions with open participation. **Interview focus**: Desired program and platform support, coordination, and remaining gaps (RQ2).	ML1: 1 excerpt; ML3: 4 excerpts; ML4: 1 excerpt; ML5: 1 excerpt; ML6: 1 excerpt; ML7: 1 excerpt; ML8: 3 excerpts; ML10: 2 excerpts; ML11: 3 excerpts; ML13: 5 excerpts; ML14: 10 excerpts; ML15: 4 excerpts. **[Total: 36 / 292 excerpts]**

II. Screening Questionnaire

This section presents the screening questionnaire for GSoC mentors and organization administrators (Org Admins) interested in our interview. We administered the questionnaire through Google Forms. It automatically collected each respondent's email address for scheduling interviews.

- **[GSoC roles and participating cycles]** What was your participation role during GSoC 2024, 2025, and 2026? Select all that apply.
  - Mentor: checkbox for each applicable cycle (2024, 2025, and 2026)
  - Org Admin: checkbox for each applicable cycle (2024, 2025, and 2026)
- **[OSS mentoring tenure]** How long have you served as a mentor in OSS, either informally or formally, through channels such as GSoC or Outreachy?
  - Less than 1 year
  - 1–2 years
  - 3–5 years
  - 6–10 years
  - More than 10 years
- **[Organization size]** How would you describe the size of your primary OSS organization in GSoC?
  - Small: fewer than 10 regular contributors
  - Medium: 10–50 regular contributors
  - Large: 51–200 regular contributors
  - Very large: more than 200 regular contributors
  - I am not sure/difficult to estimate.
- **[Project category]** Which category do you think your GSoC project falls under?
  - Artificial Intelligence
  - Data
  - End User Applications
  - Infrastructure and Cloud
  - Media
  - Operating Systems
  - Programming Languages
  - Science and Medicine
  - Security
  - Social and Communication
  - Web
  - I am not sure
  - Other: (free text).
- **[GenAI sentiment]** Please indicate your level of agreement with the following statement: *"Overall, I feel positive about the use of AI tools in OSS development."*
  - Strongly disagree
  - Disagree
  - Neither agree nor disagree
  - Agree
  - Strongly agree

III. Additional Interview Participant Demographics

Supplemental Table II extends the demographic summary in Table I with participating cycles and project categories, which show the variation in recent GSoC experience (2024–2026) and project context that we considered alongside the four primary sampling dimensions.

TABLE II
GSoC cycles and project categories of interview participants.

ID	GSoC Cycles	Project Categories
P1	2026	End User Applications, Social and Communication
P2	2024, 2025, 2026	3D Modeling Architecture
P3	2024, 2025, 2026	End User Applications, Operating Systems
P4	2024, 2025, 2026	Infrastructure and Cloud, Operating Systems
P5	2026	Data
P6	2024, 2025, 2026	Relational Database Management Systems
P7	2024, 2025	End User Applications, Operating Systems
P8	2024, 2025, 2026	Programming Languages
P9	2026	Science and Medicine
P10	2024, 2025, 2026	End User Applications
P11	2026	End User Applications, Social and Communication
P12	2026	Development Tools, Infrastructure and Cloud
P13	2026	Science and Medicine
P14	2024, 2025, 2026	End User Applications, Social and Communication
P15	2024	Artificial Intelligence, Data, Infrastructure and Cloud, Programming Languages, Science and Medicine

IV. GSoC Mentors Interview Guide

This section presents the semi-structured interview guide for recruited participants.

A. Introduction and Consent

[Interviewer introduction, contact confirmation, interview logistics, and audio-recording consent procedures omitted.]

B. Demographic Information

- **[Gender identity]** Could you tell me your gender identity? You are welcome to decline to answer.
- **[OSS experience]** How many years have you been involved in OSS development?
- **[Mentoring experience]** How many years have you been mentoring in OSS projects?
- **[GSoC experience]** How long have you participated in GSoC as a mentor?
- **[Project size]** Could you briefly describe your project's size?
- **[GenAI use]** How often do you use GenAI tools in your software development work, such as coding, prototyping, testing, design, documentation, or analysis? If we use a Likert scale: never, rarely, sometimes, often, or very often.

C. Warm-up Questions

- **[Role]** Could you briefly describe your role in GSoC and how long you have been mentoring or maintaining this project?
- **[Motivation and expectations]** What originally motivated you or your project to participate in GSoC, and what were your initial expectations?
- **[Perceived contributions]** Looking back, what role do you feel GSoC has played in shaping you, your project, and mentees?

D. GSoC Impact Questions

Thinking about how contributors join your project and become core members in your project, please answer the following:

- **[Onboarding]** How has participating in GSoC changed the way your project onboards newcomers?
- **[Collaboration]** How has GSoC reshaped collaboration norms and the way your project's contributors communicate with newcomers?

- **[Established practices]** Have any onboarding practices that emerged during GSoC cycles become institutionalized in your project (templates, good-first-issues, mentorship pairings, docs)?
- **[Contribution acceptance]** How has GSOC participation changed how you have adjusted the PR/Issues acceptance process in your project?
- **[Knowledge sharing]** How has GSOC participation changed how you have adjusted the sharing and documentation of project knowledge and expertise?
- **[Mentoring recognition]** Does participating in GSoC change how mentoring efforts are recognized or acknowledged within the project?

E. GenAI Impact Questions

- **[Initial comparison]** Has anything about GSoC felt different to you this year compared with earlier cycles because of AI?

The following questions are about AI. Thinking about how AI is reshaping open source and GSoC mentoring, compare your experience this year to previous years:

- **[Onboarding]** How has AI changed how your project onboards newcomers?
- **[Collaboration]** How has AI changed collaboration norms and the way your project's contributors communicate with newcomers?
- **[Onboarding practices]** How has AI changed onboarding practices that are core to GSoC cycles (templates, good-first-issues, mentorship pairings, docs)? Have these lost their effectiveness, been reinforced, or been redesigned?
- **[Contribution acceptance]** How has AI changed the PR or issue acceptance process in your project?
- **[Knowledge sharing]** How has AI changed the sharing and documentation of project knowledge and expertise?
- **[Mentoring role and effort]** How has AI changed what you as mentors are doing, the amount of effort you invest, and the overall impact of your role?

F. Response and Adaptation Strategies

- **[Enacted and planned responses]** What changes has your project made in response to AI or plans to make? (e.g., gatekeeping requirements, proposal templates, mandatory PRs, live interviews, AI-use policies, blocklists.)
- **[Practices from other projects]** Are there strategies you have seen in other projects or organizations that you wish your project could adopt?
- **[GenAI-assisted responses]** Are you using AI itself as part of your response because of AI, for example, AI-assisted PR review, screening, or first-pass triage? And how do you feel about it?
- **[Desired GSoC support]** If you could change one thing about how GSoC handles AI at the program level, what would it be?
- **[Desired platform support]** If you could change one thing about how GitHub handles AI, what would it be?
- **[Evolving perspectives]** How have your overall experiences with AI shifted as you have lived through these cycles?

G. Closing

- Is there anything about the impact of GSoC on you or your project, or about how AI is changing that impact, that you would like to add?

V. Interview Analysis Codebooks

A. How does GenAI affect mentoring across GSoC's pipeline, and what does it mean for the program's ability to renew itself? (RQ1)

TABLE III: GenAI effects on GSoC mentoring and example interview excerpts.

ID	Effect	Example Excerpts
GenAI-related Challenges (CH)		
CH1	Application and contribution overload	*"There is a very huge wave that comes [3 or 4 months] before GSOC... like a single contributor is making 5 PRs... so it gets very overwhelming for reviewers."* [P1]
CH2	Good-first-issues lose capability signal	*"They just AI it up, complete it, and say yeah, I understood everything. Even though they probably didn't."* [P15]
CH3	Reduced effort in proposal preparation	*"A lot of AI-generated proposals [contained] 'here is the proposal you asked for'... They didn't even bother to check."* [P5]

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TABLE III: GenAI effects on GSoC mentoring and example interview excerpts (continued).

ID	Effect	Example Excerpts
CH4	Applicant capability uncertainty	*"[The organization] accepted a contributor... [then] found out that this person does not know how to use Git... [or] run shell commands. But their proposal looked correct..."* [P4] *"AI is meaning that those tasks [of different difficulty levels] are less of a differentiator and we need to meet these people in person more often, which takes more time"* [P8]
CH5	Suspected application fraud	*"Easy access to AI has made it easier for applicants to abuse the application process"* [P6]
CH6	Silent mentees	*"Before [AI], people would ask many more questions while progressing to a solution... review went much better because you had an understanding of what the person was thinking. Now you have to be more guarded."* [P14] *"It has become very easy for [newcomers] to raise PRs and fix the bugs... But we asked them, why did you make this change? They gave us an AI-generated response."* [P1]
CH7	GenAI dependency and deskilling concerns	*"[A mentee] told me that I won't be able to create more PRs until my AI limit renews."* [P1]
CH8	Loss of productive struggle	*"Earlier, [a contributor] would come to me with a problem, and I didn't know the solution either, so we both would deep dive... There was serious mental effort... [that] helped me grow along with my contributor."* [P7]
CH9	Shift from mentoring to policing	*"I had to pay more attention to reminding people what the program rules were, aside from what our project-specific guidelines are."* [P6]
CH10	Contribution volume outpaces review capacity	*"If it's easier to write a submission, then it's harder to review a submission... even the mentors are humans. They can't review everything."* [P15] *"The only significant change has been the amount of effort we're putting into reviewing PRs."* [P4] *"The code PRs are so long that it's 5,000 lines of code. And they expect us to review the complete PR within 2 days."* [P1] *"If [an] AI disclosure [is made]... we run an extra set of tests... [review takes] maybe 20, 30% more."* [P4]
CH11	Trust erosion and verification limits	*"I can no longer trust that a human has actually checked what they are doing"* [P13]
CH12	Maintenance debt	*"We're having lots of incremental and small bits and modules [that were] very costly to maintain"* [P8]
CH13	Code reliability and security risks	*"When we give AI a prompt and it gets something working and we just push it, it can be disastrous... you never know what AI missed; it could be a security threat."* [P2]
CH14	Unreliable GenAI-assisted review	*"The AI review pointed out syntax-related issues [such as] unused functions or cleanups... but not necessarily in terms of the business logic or the mathematical [logic] because they can get complex."* [P3]
CH15	Unclear evaluation for GenAI-policy violations	*"Every other week I see a thread about whether I should fail this contributor because they did not follow my LLM guidelines"* [P12]
CH16	Weaker project attachment and retention	*"I am worried [about] how much [contributors] are learning and [whether they will] stick to the project and the community"* [P13] *"Folks drop off because they get automated replies or generic responses"* [P11]
CH17	Licensing and intellectual property risks	*"LLMs are trained on data that is not attributable... we might be violating license agreements with other open source projects."* [P7]
CH18	Policy fragmentation	*"Some [projects] have an explicit policy saying we do not accept AI contributions. Most have an AI disclosure policy... [others say] it's okay if you use AI as long as... the quality is high enough that it doesn't look AI-generated."* [P4]
CH19	Privacy and data-control risks	*"People are moving to self-hosted [forges] because they may not want AI trained on their code"* [P14]

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TABLE III: GenAI effects on GSoC mentoring and example interview excerpts (continued).

ID	Effect	Example Excerpts
CH20	Mentor verification fatigue and withdrawal	*"It's a drain on their time... [Organizations] want new people to join and grow, not... prompt a tool [and transfer review work]."* [P14] *"[AI disclosure] causes extra work for mentors... we have to stay up to date [on] whether they're following the rules and look for cues [but] that's necessary."* [P12] *"I've been a lot more worried on the impact it has on mentors. It does create fatigue... implication in terms of workload."* [P10] *"The overall amount of effort had gone up exponentially to something we can't keep up with... We have not participated in GSoC 2026. I don't think we will be participating again... It is just not in our bandwidth."* [P7]
GenAI-related Opportunities (OP)		
OP1	GenAI-reduced onboarding friction	*"The contributor was facing issues while setting up the project... AI was a huge help for me as a mentor because I was able to quickly debug and answer their questions"* [P2]
OP2	GenAI as a mentoring multiplier	*"If I had mentored three or four years ago, I would have struggled to manage the time"* [P11]
OP3	GenAI pre-review of contributions	*"GitHub Copilot first reviews [a pull request]... and they add some comments... common [issues] are detected automatically, and the contributor can immediately get feedback... without waiting for a human."* [P5]

TABLE IV: GSoC sustainability functions, definitions, and example interview excerpts.

ID	GSoC sustainability	Definition	Example Excerpt
IM1	Talent recruitment and entry	GSoC helps OSS organizations identify and recruit motivated, capable newcomers.	*"We got most of the contributors from [GSoC]... this has been one of those hot spots [for newcomers]."* [P2]
IM2	Developing trustworthy community members	GSoC develops contributors trusted with responsibility through demonstrated understanding and commitment.	*"We are judging the students even before they apply... [based on] six months contributi[on] for us... even if [they] does an AI-generated application, we know [they are] capable."* [P3]
IM3	Skills development and career signal	GSoC develops technical skills, judgment, and professional competence and signals career readiness.	*"GSoC has been pretty monumental... helped me secure Linux Foundation opportunities, grants... and the technical competence to see them through."* [P12]
IM4	Maintainability of funded contributions	GSoC funds verified, maintainable improvements to under-resourced OSS projects.	*"GSoC has attracted many contributors we would not normally have gotten... helped us solve longstanding issues nobody was in the mood of working on."* [P13]
IM5	Institutionalization of governance practices	GSoC establishes lasting organizational routines for OSS mentoring and governance.	*"All the decisions we make, we write it down... we won't repeat the same thing again and again every year"* [P1]
IM6	Community growth, diversity and vitality	GSoC builds a diverse, lasting contributor community through sustained newcomer engagement.	*"GSoC gives people who are new to open source a chance to contribute and hopefully keep contributing... the aim is mostly keeping the organization alive and thriving."* [P14]
IM7	Contributor-to-mentor renewal	GSoC renews OSS communities by developing contributors into committed maintainers and future mentors.	*"Almost 70% of [2026 mentors] had been our previous students... we are growing mentors who can guide the next generation of contributors."* [P3]
IM8	Mentor motivation and retention	GSoC sustains mentor participation through personal and community benefits and manageable workloads.	*"GSoC has had a huge impact on my career... [and] gave me a chance to mentor.. I've always been on the receiving end of [GSoC]... helping someone else was a great experience for me."* [P15]

B. How do mentors and organizations respond, and what support do they expect from GSoC? (RQ2)

TABLE V: Strategies for responding to GenAI in GSoC and example interview excerpts.

ID	Strategy	Excerpts
ST1	Selecting applicants by prior contribution trajectory	*"We are judging the students even before they apply... [based on] six months contributi[on] for us... even if [they] does an AI-generated application, we know [they are] capable."* [P3] *"One thing we looked out for this year was a consistent history of high quality contributions prior to GSoC... all of [the selected candidates] had very good PRs already merged... interactions genuine enough to make us trust... they were reviewing their own code."* [P4]
ST2	Live interviews to verify applicants' unaided ability	*"We do not go by the applications. We try to take face-to-face interviews, video interviews with screen sharing on... scenario-based questions... [and] make them write the code in front of us in the editor"* [P3] *"Next years... we will try to conduct interviews before selecting... are they really strong enough without AI?"* [P1]
ST3	Entry tasks that require project-specific understanding	*"[We need tasks requiring] more cognitive thought and understanding of our processes... so that just a prompt with the task will not be able to produce the output"* [P8]
ST4	Rewarding honest, rule-following GenAI use in selection	*"[Mentors] did try not to select folks who were using AI, not because of a refusal to accept any AI code, but... to compensate for the abundance of applicants... using AI without paying attention to rules."* [P6] *"Candidates... looked eligible... but wherever we had some hint [that] they might be using LLMs and were not honest... we put them on a lower priority list."* [P12] *"Some of [AI-generated proposals] came back and rewrote it. And I was like, okay, this is fine."* [P9] *"We took the effort of manually going through each proposal [in a filtering session]... marked AI slop proposals as not fit... stack-ranked all the non-slop proposals [and] gave a chance to improve."* [P11]
ST5	Banning GenAI for specific outputs with explicit exceptions	*"I used AI for documenting the project... AI is good for documentation... I trust AI on this."* [P5] *"We specifically asked in our project description: please do not submit any code using AI."* [P5] *"I have a strict ban on artificial intelligence use"* [P7] *"You may not use AI or LLMs in any way"* [P9]
ST6	Allowing AI conditionally: "assistant, not owner"	*"We do follow a practice that always recommend to use AI as long as you know what you are working on and how it is working"* [P2] *"[AI] is a tool... [mentees may] check grammar, bugs, and style... We allow [that], and that's fine"* [P12] *"You are using AI as your assistant. AI is not your owner"* [P1] *"You control the AI; you don't let the AI control you"* [P2]
ST7	Judging quality, not authorship	*"Whether it has been generated by AI, whether it has been human-written... if things are working fine, then people are happy with it."* [P3] *"Code quality is more important than flagging someone for using AI."* [P15]
ST8	Using GenAI for mentoring support	*"We do have GitHub Copilot [and] use that on occasion to review larger changes... It is much better than us at catching things [like] security, subtle bugs, edge cases."* [P4] *"One was I quickly got AI to review the whole PR code base... the second thing, I myself reviewed the PR"* [P2]
ST9	Teaching contributors to use GenAI responsibly	*"We gave [contributors] written instructions on what we are expecting with the use of AI, and we encourage them to use it... [but] they shouldn't be completely dependent upon it."* [P2] *"You can use my expertise to help you to understand how to check things, but I am not being your checker... My role is to mentor you to develop your skills."* [P8] *"If it breaks, you need to know how to identify and fix the issue"* [P15]
ST10	Keeping final decisions with humans	*"Even after AI said it was good to go, then we have to manually review it."* [P5]
ST11	Treating GenAI-suspicion cues as prompts for closer review	*"It is next to impossible to confidently attribute something [as] AI-generated... we need to look for heuristics. We can't really depend on hard and fast rules. But when we do find definitive proof... we swiftly ban users from our repos."* [P7]

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TABLE V: Strategies for responding to GenAI in GSoC and example interview excerpts (continued).

ID	Strategy	Excerpts
ST12	Asking contributors to explain their code in review	*"I'll ask little things like, 'Why did you choose to do it this way?' If they can't answer, then... [it is] a sign [of AI use]. But I don't accuse them immediately... It's through a discussion, 'How did you check it?', that you can have a learning relationship."* [P8] *"If [the contributor uses] AI properly, understands the logic, and makes you understand it, it will be almost impossible to tell whether AI [was] used."* [P3] *"We sometimes ask them to clarify their prompt. If they use AI right, we probably won't see it; if they use it to do their job instead of themselves for them, we will. We don't have a philosophical rejection of AI."* [P10]
ST13	Requiring contributors to disclose GenAI use	*"It would be better if [the organization] put [AI disclosure] as a strict guideline so mentors don't have to do... guessing work... mentees [should] explicitly list whether and how they use LLMs."* [P12]
ST14	Coordinating GenAI guidance across the program	*"[GSoC could] come up with a guideline [for AI usage]... but each and every organization is differently structured. They follow their own rules along with the GSoC rules... generalized practices could be spread along."* [P2] *"At the program level... require a strict form where the mentee discloses in what capacity they're using AI... [so that] eases the burden from each community individually enforcing those rules"* [P12] *"[At the program level,] having an AI... scan proposals... and flag it if they think it's... LLM-generated. Not to say that we would immediately disregard that person, but just as an external flag so that we don't have to... unpick everything... we suspect this might be LLM-generated, and then you can look through with that lens."* [P8]